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Leveraging insights from neuroscience to build adaptive artificial intelligence

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Mackenzie Weygandt Mathis

Biological intelligence is inherently adaptive—animals continually adjust their actions in response to environmental feedback. However, creating adaptive artificial intelligence (AI) remains a major challenge. The next frontier is to go beyond traditional AI to develop 'adaptive intelligence', defined here as harnessing insights from biological intelligence to build agents that can learn online, generalize and rapidly adapt to changes in their environment. Recent advances in neuroscience offer inspiration through studies that increasingly focus on how animals naturally learn and adapt their models of the world. This Perspective reviews the behavioral and neural foundations of adaptive biological intelligence, examines parallel progress in AI, and explores brain-inspired approaches for building more adaptive algorithms.

Our world is built on a series of predictions that we make in our minds. That is, to us, the world is a construct, conceptualized from a series of millions of predictions harbored in the neural code. When we walk on the beach, we accumulate photons with our eyes, we detect sound waves with our ears, we feel the wind on our face and the sand shifting under our feet, and all of this comes together to give us the perception of a warm, sunny day with ocean waves crashing on the shore. However, our brain sits in the dark confines of our skull, and tens of billions of neurons communicate through electrical activity in a bath of chemicals. As incoming sensory information is processed in the brain at different times, the brain imposes a lag to compile it, thrusting our perception into the past, tens of precious milliseconds behind reality. Nevertheless, we need to act quickly, and our brains have evidently evolved solutions to overcome these delays.

Moreover, we need to adapt to environments marked by unpredictability, continuous change and infrequent repetition of decision-making scenarios. The uncertainty, variability and complexity of these settings demand adaptive strategies, crucial for learning about local changes—such as the shifting sand under our feet. Dating back decades, researchers have developed new frameworks to understand (decompile) how this global prediction system may work[1–4]. A core tenet has been that, to overcome sensory-processing delays, we must build internal models of the world (also called world models). We use the models to predict the sensory and motor consequences of our actions[5–8]. However, it remains largely unknown how the brain enables adaptive behavior, and it is of growing interest how we can take inspiration from the brain to build more adaptive AI[9–11]. AI here broadly refers to systems or machines that aim to mimic biological intelligence by performing tasks that typically require cognition, such as problem-solving, decision-making, learning and language processing. Technically, it encompasses machine learning (ML), deep learning and foundation models in fields such as natural language processing and computer vision. In this Perspective, I review progress in the study of biological intelligence and use it to propose ways to enhance AI-based agentic systems with internal models.

Internal models for adaptive behavior

Dating back at least as far as Aristotle, there has been the philosophical and scientific question of how our brains build models of the world[12]. How can we combine sensory information into perception? How long does this take? Are we always tens or even hundreds of milliseconds behind reality? How do our brains make predictions about the world in order to act faster? Why do we mostly notice when the world violates our predictions? If the ocean were suddenly quiet, we would immediately worry that we had lost our hearing, rather than suspect that the ocean had completely changed. While this is familiar in everyday life, we need to turn this experience into a controllable task to study it in the laboratory.

What is the neural basis of this adaptive behavior? This has historically been difficult to answer because we are often undersampling the neural data that underlie the behavior, and such sparsity frequently requires repeated measures for statistical robustness, which is not

École Polytechnique Fédérale de Lausanne (EPFL), Brain Mind Institute, Geneva, Switzerland. e-mail: mackenzie.mathis@epfl.ch

ideal for studying changing behaviors. Moreover, the mouse brain has an estimated 70 million neurons, yet even the most state-of-the-art technology allows for recording up to 1 million individual neurons at a time[13] (and on the order of only 3 Hz resolution), while most labs can record in the order of hundreds to thousands of neurons with two-photon microscopy or high-density arrays such as Neuropixels[14,15]. Thus, classically, to study how a neuron encodes a given stimulus, the animal is repeatedly subjected to the stimulus, and then we build encoding and decoding models[16] to gain insight into what the neural activity represents.

However, if you want to understand how the brain learns and updates internal models, there is an inherent problem with this repeated-trial approach. As Heraclitus infamously quipped, "No man ever steps in the same river twice, for it is not the same river and he is not the same man." Beyond the clear changes that can occur across trials during learning, the neural code can change even in a steady state, given that the output behavior is rarely truly identical. Thus, averaging across trials likely misses key principles of neural computation. Since behavior is rarely exactly repeated, investigating learning—where behavior changes in predictable and measurable ways due to task design—may offer crucial insights into the neural basis of adaptive behavior. Namely, we can control the environmental factors that drive learning and examine the neural code across time. This not only lends itself to more ecologically relevant tasks (the animals must be able to learn), but it is also becoming a tractable approach given modern technological advances in large-scale neural recording and joint neural and behavioral analysis[16–19].

Adaptive animal and neural behavior

There has been a concurrent push for experimental designs that incorporate more naturalistic and ecological behaviors, and tasks that enable within-session learning. Aside from new large-scale neural recording technology, a part of this expanded task space is due to the advances in markerless tracking of behavior[20], which has made digitizing complex movements much more tractable. What behavioral paradigms are being developed? Although not all progress in the field can be covered here, several new approaches are highlighted. Freely moving paradigms for rodents have rapidly advanced to encompass complex stimuli such as moving (threatening) new objects[21], real-time virtual reality worlds[22], labyrinth mazes[23], extended multi-area home cages[24,25] and 3D environments such as those used to study depth estimation[26]. In several settings, they have also built trial-based assays that smartly constrain naturalistic settings to directly probe learning across epochs.

An elegant thread of work across multiple groups aims to build tasks for zero- or few-shot learning in complex spatial worlds. Zero-shot learning is a term adapted from ML that describes the ability of a model to generalize to new tasks or concepts without any specific training examples[27]. Few-shot means that it can learn from only a few examples. For example, ref. 23 reports the development of a task where mice must seek rewards in a labyrinth (Fig. 1a). They found that mice cannot only make up to 2,000 decisions per hour, but after unsupervised exploration of the maze, they could few-shot learn to find the water spout (in around ten tries)[23]. Notably, this is a learning rate that is 1,000 times higher than those typically observed in two-alternative forced choice tasks deployed in rodent studies of decision-making, meaning this task allows a more thorough study of rapid learning. Another example is showing that mice can zero-shot learn new sequences of rewards (such as go to location A, B, D then C)[28]. Mice were trained on multiple tasks with a shared underlying structure that organized a sequence of goals, although the specific goal locations varied. Strikingly, the mice learned this common structure, allowing them to make zero-shot inferences on the first trial of new tasks (Fig. 1b). Intriguingly, they also found neurons in the medial prefrontal cortex that acted as task-structured memory buffers, namely, they tracked the progression along behaviorally relevant steps[28]. Such schemata could perhaps be mapped to memory replay in neural networks (see below), and to internal models.

Animal intelligence is, of course, not limited to laboratory animals. Dating back to early comparative cognition studies[29], there has been a large body of work studying intelligence in a myriad of animals. Crows, for example, exhibit tool use and causal reasoning, with New Caledonian crows even manufacturing tools to retrieve food[30,31]. Bees demonstrate numeracy and complex communication through the waggle dance, encoding spatial information about food sources[32], and can solve 'puzzles' such as pulling a string to move a food source closer[33]. Various domesticated animals also exhibit human-aligned intelligence. Horses show social learning and can interpret human emotional cues[34]. Dogs display theory of mind and word comprehension that rival those of young children[35]. Such findings even sparked a recurring virtual animal AI Olympics challenge[36].

In sensorimotor control, there has been a long line of research on rapid learning (and updating of internal models), commonly referred to as motor adaptation. For clarity, learning is defined as the acquisition of a skill, which involves forming a new internal model, whereas adaptation refers to the use of a learned internal model and its subsequent updating. Motor adaptation studies have been developed in which visual or proprioceptive information is perturbed, requiring an animal to adapt within a single session. This has spurred the development of not only new neural analysis tools[16,37–39], but also a host of behavioral tasks. For example, visuomotor rotations have been used in humans and non-human primate studies to study how they can account for sensorimotor discrepancies in only a few hundred trials[37,40–43]. The same principle is used to change the environmental dynamics of a manipulandum, which causes deviations in limb-movement tasks[7,44–46]. Notably, these tasks are specifically designed to measure both adaptive learning and the formation of internal models. There is typically a period of baseline control movements, perturbations, and a return to baseline conditions that allows researchers to behaviorally measure whether an internal model was updated (Fig. 1c).

Neurally, evidence shows that motor and sensory areas can change their tuning properties during the course of learning[46–50]. A study in ref. 48 showed that during a motor adaptation (learning) task in macaques, motor cortex (M1) can 'index' memories of the hand-force required for the process of learning to adapt in the form of movement readiness potentials that occur before movement execution. Notably, the neural subspace most predictive of hand forces changed during the period before movement (preparatory), specifically during the learning epoch. In a neural dimension orthogonal to this force-predictive subspace, the researchers identified a uniform shift across all movement directions, including those unaffected by learning. Notably, this uniform shift persisted even after exposure to the force field, suggesting an updated internal model. In addition, other works show new evidence of state changes that persist across learning[8,46], and temporally resolved prediction errors[46,51]. Influential work on visual cortex (V1) has shown the existence of 'mismatch' neurons when the expected visual feedback is disrupted[51], and similar prediction errors have been found in somatosensory (S1), motor (M1), and frontal areas of cortex[46,50].

Some of the best evidence causally showing how neurons adapt has come from brain–machine interface (BMI) studies that require the participant to directly alter neural firing to change something in the external world (like a cursor on a screen). Closed-loop systems, such as calcium-based BMIs (BMIs driven by decoded optical activity readout of calcium fluorescence signals) or those with electrical stimulation, have been used to study the timescale and subsets of neurons that can be used, as not all neuronal types have been found to be equally adaptable. Fundamental work has revealed that even small numbers of neurons can be leveraged for decoding and for BMI-guided

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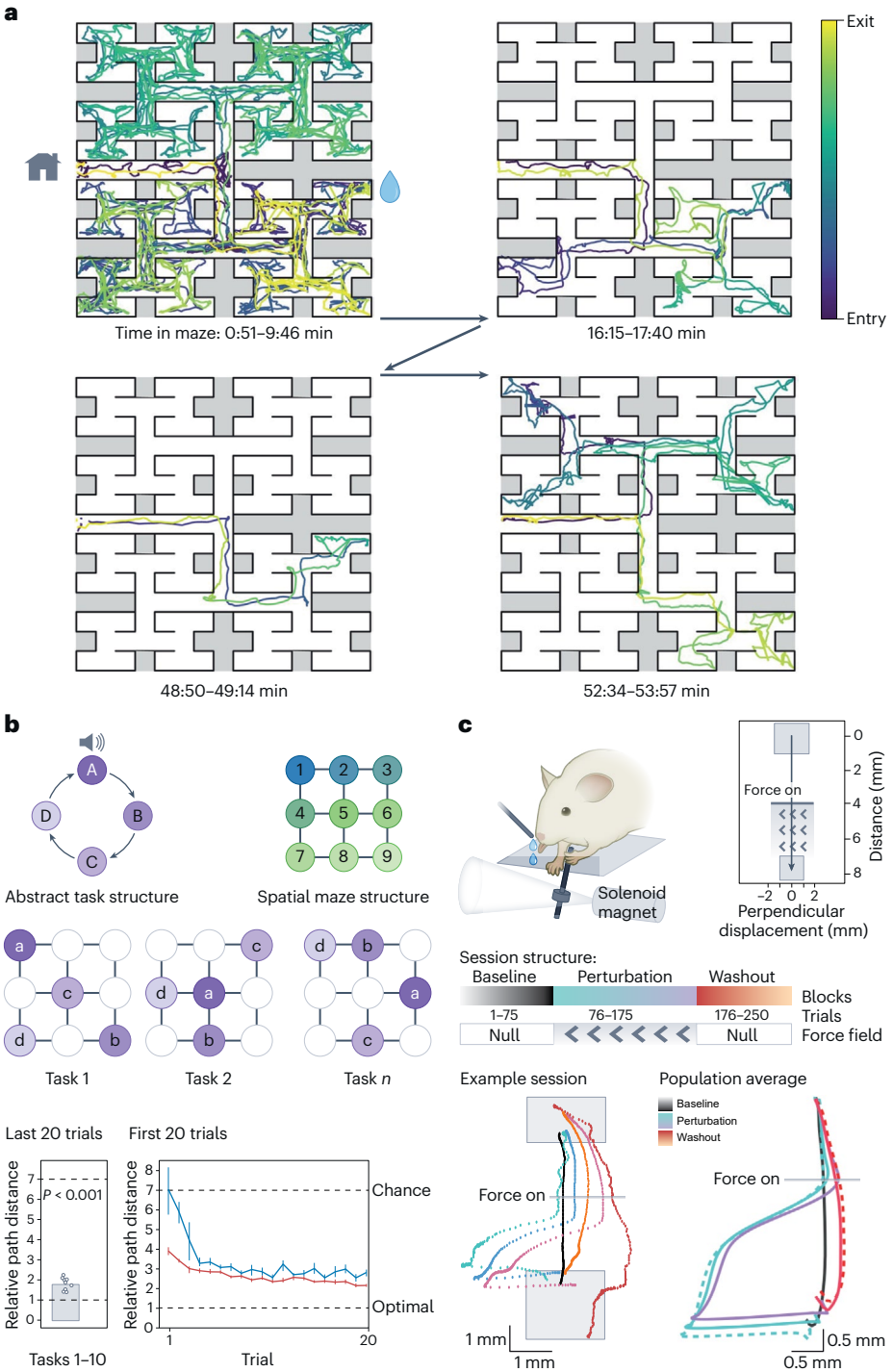

Fig. 1 | Rapid learning in animals—from few-shot to updating of internal model-based learning. a, Adapted from ref. 23, CCBY 4.0: four sample bouts from one mouse (B3) into the maze at various times during the experiment (time markings at bottom). The trajectory of the animal's nose is shown; time is encoded by the color of the trace. Top left, the entrance from the home cage and the water port are indicated. **b**, Adapted from ref. 28, CCBY 4.0: animals learned to navigate four sequential goals on a 3 × 3 grid-maze. Reward locations varied, but the ABCD loop structure remained constant. With multisession learning, they reached near-optimal paths in the last 20 trials. Performance also improved in the first 20 trials of each task, with faster gains in the last five tasks than in the first five. **c**, Adapted with permission from ref. 7, Elsevier: diagram of the task, consisting of 75 baseline, 100 perturbation and 75 washout trials. Example session shows baseline (black), first perturbation (sea green), early (green) and late (purple) perturbation trials, and early washout (orange/red). Across 27 sessions, aligned averages show early versus late adaptation and washout, with colors matching the single-session plot.

feedback[37,52]. Notably, they often live in discrete subspaces of the neural dynamics. Ref. 53 further extended this by linking these subspaces to specific cell types. The authors trained mice to modulate the activity of either intratelencephalic or pyramidal tract neurons in response to reward. They found that mice learned to control pyramidal tract neuron activity more quickly and effectively than intratelencephalic neuron activity. This could be related to the anatomical connectivity and differing inputs.

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Anatomical links to computations

The anatomy of the brain is important to consider, as this too could have direct implications for the future design of adaptive AI systems. Several areas in the brain have this remarkable structure, layers in the cerebral (neo-) and cerebellar cortex, which could be a critical part of new architectures for AI systems. There is a long-standing debate about the emergence of representations from data versus architectural constraints[54,55], but it is increasingly clear that architecture is critically important and linked to function. Such architectural bias gives rise to computations on information that incorporates brain-area-specific information[56]. Notably, the architecture arises during development, and such inductive biases from evolution are interesting to consider (Box 1).

Form follows function, and one emerging hot topic is how cortical circuits implement learning from prediction errors across different cell types and layers. Building on the initial discovery of sensory prediction errors in cortex[51], subsequent works have begun to record from subtypes of neurons such as parvalbumin, vasoactive intestinal peptide (VIP) and somatostatin classes of interneurons and from excitatory neurons to form a more complete model of how both excitatory and inhibitory neurons across layers could implement the learning rule[57–59] (Fig. 2a). This work in cortex follows earlier work that discovered reward prediction errors (RPEs) in the midbrain (within the ventral tegmental area)[60–62] and found that they are also part of an inhibitory-gated circuit wherein GABAergic neurons have a crucial role in the computation of RPEs[63] (Fig. 2b).

Recent work aims to unify how a hierarchical implementation of prediction errors (including RPEs) could aid in learning across sensory, cognitive and motor systems. Ref. 64 found that during a cue-guided reward task, layer 2/3 somatosensory neurons showed enhanced responses to reward-predictive stimuli, and this led to reduced RPEs and increased confidence in predictions. Following rule reversal, the lateral orbitofrontal cortex, via VIP interneurons, signaled context-prediction errors, reflecting a loss of confidence. The work suggests a hierarchical interaction of prediction errors across cortical regions, with top-down signals modulating the activity of the sensory cortex. Notably, previous work finds limited roles for reward-only-driven motor learning, but it can be combined with sensory prediction errors to shape performance[7,42,65].

Measuring changes in the neural code

New tools are critical for the study of adaptive behavior. Computational neuroscience has made large advances in the past decade, fueled by the need to better understand ensembles of neurons[18,66–68] and by leveraging the concurrent innovations in deep learning, which is reviewed in refs. 16,69 In brief, algorithms such as variational autoencoders, transformers, contrastive learning-optimized neural networks and diffusion models are already making big strides in our ability to combine data across participants, model multiregion interactions and 'decode the brain'[16]. Several tools enabling time-series analysis in single neurons and neural populations are highlighted, as they are critical for adaptive learning studies and are likely to become increasingly popular for studying AI systems.

One of the most impactful tools in the last 20 years has been the adoption of generalized linear models (GLMs) and generalized additive models for neuroscientific applications[70–72]. Here a model is trained using a feature matrix (which can contain one or many features) with the task of predicting the spikes of neurons given the set of features. While GLMs can come in many flavors, as an example, generally to handle variability in behavior, models are trained and tested on different splits of the data (also for cross-validation). Predictions from these test sets are combined, and a pseudo-R^2 is calculated. This requires that behavior is minimally repeated and also allows for flexibility, meaning there is no direct trial averaging or behavior-triggered averaging of the neural code.

BOX 1

Neural development and inductive biases

In the context of biological learning and the links to AI, another critical angle is neural development. There are natural connections to make that could also inspire new inductive biases in the form of architectures or optimization algorithms in AI systems. For example, during development, cells in the cortex migrate to their final resting location at different times[121]. This process of spatial and temporal organization, I believe, has parallels in AI, particularly in how artificial neural networks undergo initialization and optimization processes to organize their nodes across layers in deep learning models. Moreover, synapses, such as those at the neuromuscular junction of α motor neurons and muscles, are formed after activity-dependent pruning[122]. This mirrors the pruning techniques (also known as knowledge distillation in ML[123]) used to improve model performance, wherein unnecessary connections are discarded during training to enhance efficiency.

There is also a complex and highly regulated developmental program in the spinal cord, involving transcription factors and Hox genes[124]. This genetic regulation outlines global (rostral–caudal) and local (dorsal–ventral) connections. Similarly, the initial structure of a neural network is determined by a set of weights and biases, which are refined through training to create the connections necessary for learning. Additionally, the so-called genomic bottleneck has a key role in determining early innate abilities of animals, from the immediate suckling reflex to the more time-intensive but impressive ability of a horse to stand and walk a few hours after birth[125]. In AI, this is akin to pretrained models or transfer learning, where networks leverage previously learned features to rapidly adapt to new tasks with minimal input.

Moreover, in the cortex, early traveling spontaneous waves shape the basis of early connections between cortical neurons. These waves are believed to synchronize neural activity across various brain regions. Intriguingly, new work has highlighted how the transformer architecture naturally gives rise to waves, which is a critical feature in the brain[126]. This is reminiscent of how attention mechanisms in transformers enable dynamic, wave-like information flow across different parts of the model, helping the system maintain coherence across large datasets. Waves are observed not only in early development but also in steady state, where they are thought to maintain global connectivity in the brain, encoding states such as arousal and attention. This functional organization in the brain has direct analogs in how AI systems, like attention networks, maintain global contextual understanding across their layers.

For populations of neurons, several methods have been used or developed, ranging from principal component analysis for measuring neural trajectories[73,74] and contrastive learning paired with consistency metrics to measure changes in the latent neural dynamics across learning[39,46] to nonlinear dynamical systems for task- and task-irrelevant metrics[75,76]. These are more deeply covered in reviews elsewhere[16,69,77], but I will note that several approaches can be used for single-trial dynamics and have been especially useful in motor learning and adaptation studies in non-human primates[68,78,79].

Recent advances in AI are expected to usher in a wave of new discoveries and tools in (neuro)science. For example, large language models (LLMs) are already being used in behavioral neuroscience to digitize and quantify actions in video-based data[80]. New vision

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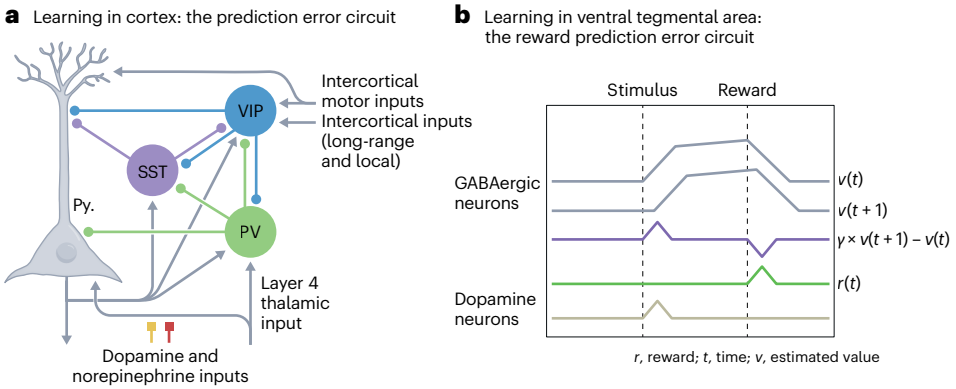

Fig. 2 | Neural computations: biological teaching signals. a, Diagram of the theoretical prediction error circuit (based on evidence from refs. 51,57,59,120). **b**, Diagram of the computation in the RPE circuit of the midbrain (ventral tegmental area) dopamine and GABAergic neurons. Inspired by refs. 61,62.

language, hierarchical, and multitask foundation models are surely coming[81]. For neural analysis, several groups are already actively working on building more unified encoders that can not only be used for brain–machine interfaces but also serve as models of the brain[16,39,79,82–84]. This is sure to give rise to new approaches to extract meaningful computational principles from neural dynamics, and these new learning rules have great potential to directly influence how we train, optimize and deploy ML systems.

Training and learning in artificial systems

Next, to understand how neuroscience is poised to influence modern AI, I provide some background on contemporary AI approaches. Today, most AI applications in production revolve around a train–test–deploy cycle and therefore are inherently not adaptive. A dataset is curated, which can be lab-project scale (or, in the case of generative pretrained transformers (GPTs), this would include nearly all the 'open-source' content of the internet), and then the data are split into a train-and-test fraction. Most efforts benefit from leaving out-of-distribution (OOD) data in the test set for a realistic measure of how well the trained model generalizes. OOD data are defined as data that are substantially different from the training data. If generalization is not needed—perhaps you are training a specific model for gait analysis in a specific laboratory setting—then the train/test split is often a random subset of the original dataset. But for many applications, training once and deploying a robust model is ideal.

How do we build robust, generalizable models that can handle changes in the real world? While the remarkable progress of GPTs in LLMs and the discovery of scaling laws have massively accelerated progress in AI, we aim for something better, more power-efficient (Box 2) and more adaptive. What has been tried? This is where, at the very least, the subfields of continual learning (lifelong learning) and in-context learning come into view.

Current approaches to adapting models

Continual learning is the task of having a neural network learn a new series of tasks of the same modality, such as computer vision tasks, over time. It aims to address the challenge of developing models that can learn incrementally from a continuous stream of data without forgetting previously acquired knowledge. This is something that biological brains excel at—we typically do not forget older information as we learn new skills. Yet traditional ML models often face catastrophic forgetting[85] when retrained on new data. However, new advances in continual learning have introduced techniques such as local module composition[86], knowledge distillation[87], elastic weight consolidation (EWC)[88,89], synaptic intelligence and memory replay[90,91] to mitigate these issues. For a more extensive discussion, see refs. 92,93.

BOX 2

Learning with spiking neural networks

One of the most biologically inspired and energy-efficient advances was the development of spiking neural networks (SNNs) in the 1990s. Called the "third generation of neural networks"[127], SNNs are unique in their ability to model a time-dependent spike process, making them not only biologically more plausible but also highly energy efficient, and can be used on edge-computing devices. Notably, SNNs naturally lend themselves to hardware-accelerated devices and neuromorphic computing. Platforms such as Intel's Loihi, IBM's TrueNorth and SpiNNaker provide hardware specifically designed to run SNNs efficiently[128]. Moreover, several early works highlight their modularity and utility for multi-area modeling[129], as well as how multi-area SNNs can be used with semantic pointers for multi-area weighting of information[130]. Historically, they have been difficult to train and scale, but that is rapidly changing, and new methods to transfer information from ANNs to SNNs[131] and/or to directly compute the required gradients from SNNs[132] are emerging.

Although not directly brain-inspired, EWC works by constraining certain parameters to reduce interference between tasks, effectively mitigating catastrophic forgetting[88]. More recent work has adapted EWC to fine-tune self-supervised models, yielding performance improvements on tasks with biased datasets by maintaining knowledge of previous tasks more effectively than traditional methods[89,94]. If we attempt to map this to neural dynamics, this might be related to the specialization of circuits. Namely, the 'parameters' of certain areas (such as primary receptors) could be fixed, whereas others (such as hippocampus and neocortex) could be more plastic.

By contrast, synaptic intelligence is a learning rule deeply inspired by brain plasticity[95]. The authors developed intelligent synapses that adaptively store information, minimizing the forgetting of previously learned tasks while acquiring new ones. This approach mimics biological neural networks, which balance plasticity and stability, thereby enabling continual learning in artificial neural networks. Notably, this method can perform as well as EWC but can be performed online. Now, with modern scalable computing and architectures, this type of learning rule could be pivotal for building more adaptive systems.

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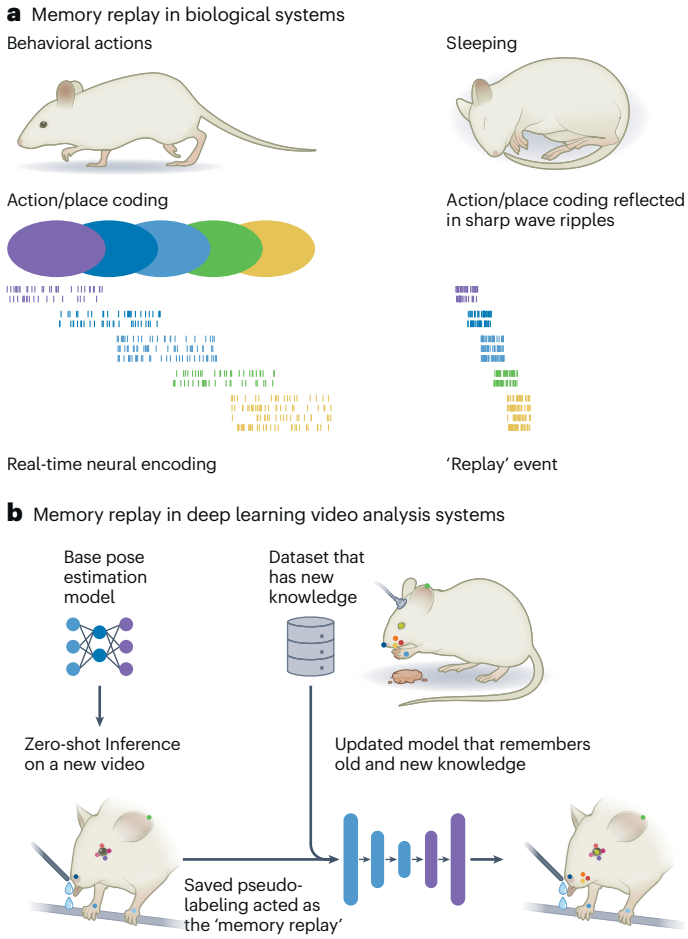

Fig. 3 | Memory replay in biological and artificial systems. a, In biological systems (such as in mice), neurons have been shown to encode place in their firing rates. These neurons can be active in the same sequence in sleep, which is thought to reflect memory replay. **b**, In deep learning systems, one can build a replay buffer in continual learning tasks. For example, in video analysis for animal pose estimation, one can use inferred labels with high confidence as pseudolabels to retain base-model knowledge, even when the base model's data might not be available to fine-tune or when a new dataset, perhaps with new labels, as depicted, needs to be learned without forgetting other labels.

Another brain-inspired ML algorithm for learning has been the development of memory replay[96,97], which mimics the memory systems in the hippocampus (Fig. 3a). Notably, memory replay also, perhaps indirectly, is built on the memory-inspired system of the Hopfield network, which is a type recurrent neural network designed for content-addressable memory, resembling a spin glass system[98], whose invention won John Hopfield a Nobel Prize in 2024. Memory replay aims to have a memory bank of already learned actions (or images or tokens, etc.) that can be 'replayed' (interweaved) into the training data to limit catastrophic forgetting (Fig. 3b). This became a popular technique in reinforcement learning and, more recently, in computer vision and in LLMs. This training approach can also be used during active learning, when new data (unlabeled, pseudolabeled or labeled) can be used to fine-tune a model. If the input data continue to morph into OOD, then using memory replay can be leveraged to ensure the prior performance can be achieved.

Several memory replay-based approaches in LLMs emerged after the release of ChatGPT in November 2022 (with its impressive language abilities yet limited content window-at-the-time, that is, lack of a long-range memory), such as AmadeusGPT[80], MemGPT[99] and Voyager[100]. AmadeusGPT introduced a short- and long-term memory module where specific keywords could be quickly recalled to overcome token limits (namely, the 4,096 in GPT-3.5), while MemGPT used a vector database to construct a persistent memory to maintain context across interactions, and Voyager iteratively refined its skills by replaying feedback from past actions in games such as Minecraft.

How to build more adaptive AI

How can we draw inspiration from adaptive behavior and the brain to develop new AI approaches[11]? A core component of adaptive behavior is having already learned priors—these aforementioned internal models—that can be dynamically called upon. These priors clearly require data at scale to form—these 'data' can be encoded into neural circuits via the genome or learned throughout life. An example of a genetic prior would be the previously metioned innate ability to suckle at birth across many species while a nongenetic learned prior would be a motor skill such as playing a musical instrument. Another data source is the massive amount of unsupervised sensory and motor stimuli we rapidly accumulate across development and life. But how do we turn data into internal models?

While many questions remain about how these models are implemented in neural dynamics, there is increasing evidence of core computations such as the previously discussed prediction errors[51,101] across brain regions, which shape and update internal models. Can we use these prediction errors as teaching signals in neural networks? Moreover, these prediction errors are anatomically defined, imposing architectural constraints on these systems that stem from brain–body dynamics. Can we leverage this feature in adaptive AI systems? Notably, these prediction errors are canonical and act in specific sensory reference frameworks (such as vision or audition). As Mountcastle noted, it is less about 'sensory' or 'motor' spaces but rather a function of the data input/output and, by extension, the shared architecture and computations[56]. Thus, it can be argued that inspiration should be drawn from building specialized nodes with core computations in the appropriate reference framework, which use prediction errors to efficiently update modules, much as the brain uses prediction errors to update modality-specific internal models.

Foundation models

Let us next consider what is currently the top approach for building generalizable models whose aim is often to ingest one or several data modalities (such as text and video) and output either. These unified so-called foundation models generalize to unseen data or can be used for transfer learning in downstream tasks[102]. More formally, these are defined as a single large-scale model trained on broad data distributions that can be adapted to a range of downstream tasks with minimal task-specific fine-tuning[102]. For example, a foundation model for natural language processing takes large-scale unlabeled text data and learns a joint representation within a model (a transformer) using tokenization and self-supervised learning[102] (Fig. 4a). The most infamous of this type is ChatGPT and other GPT models (such as the Gemini model series[103] or DeepSeek[104]). In neuroscience, several foundation models are also emerging for specific domains, such as for animal behavior analysis[91] and for modeling the visual systems of mice[84].

Other emerging and prominent approaches are generalizing this approach to truly multimodal data[105,106]. One such example is Flamingo, a vision language model (VLM) that integrates pretrained vision and language models, processing mixed visual–text data and handling images or videos as inputs[105] (Fig. 4b). This approach could be extended to more modalities, as recently done for robotics[107]. This is a rapidly advancing area of research that will surely influence neuroscience in the coming years.

Although this direction holds promise for multimodal models with enhanced generalization, efforts can be extended to develop truly smart, adaptive, and agentic systems. We should take inspiration from our brain's specialization and build a series of smartly

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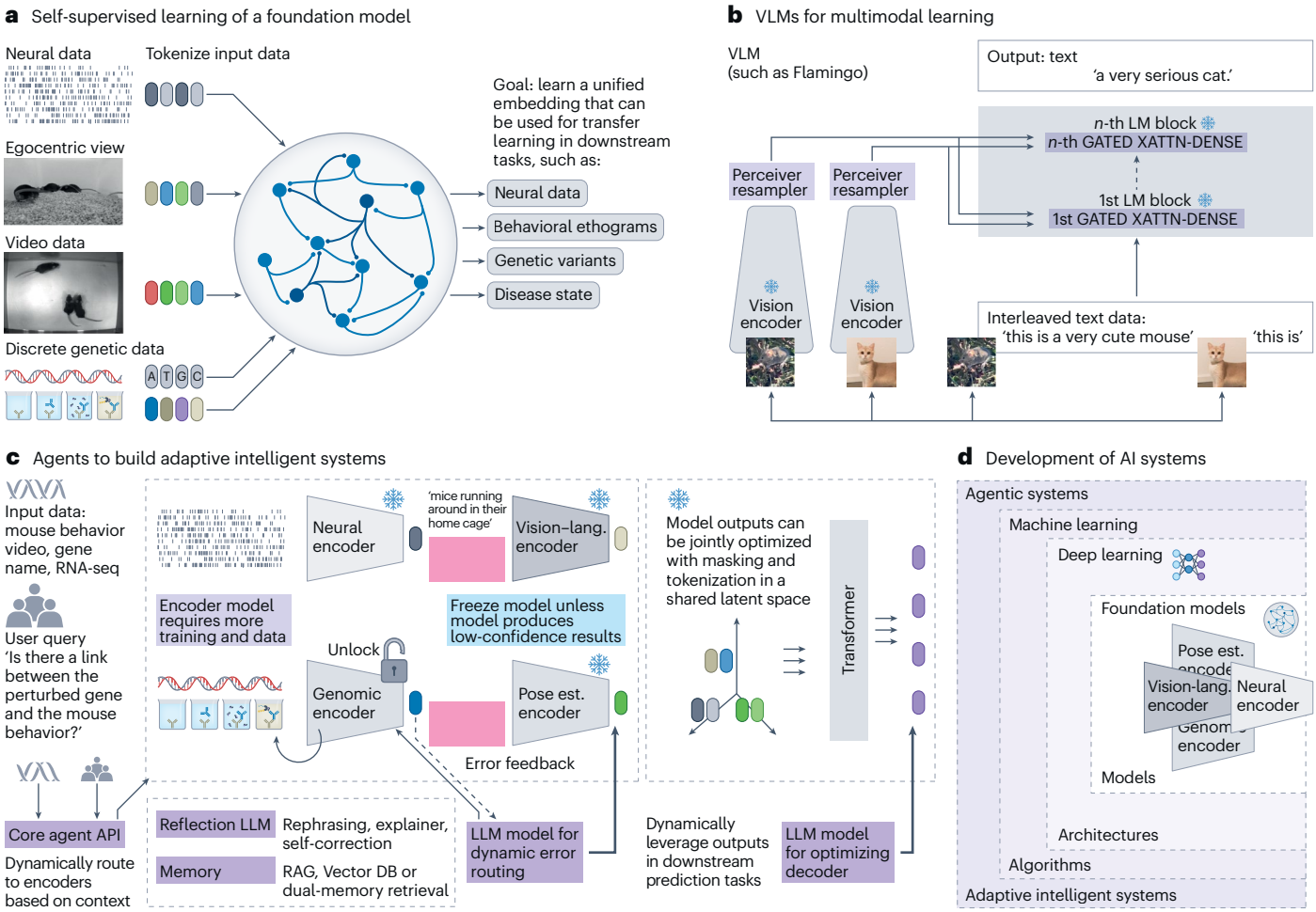

Fig. 4 | Foundation models and adaptive agents. a, One path to building generalizable models is to input diverse data, tokenize them and train collectively with self-supervised learning. **b**, Example of VLMs that interweave text and images. The inspiration and the cat image are taken from ref. 105. **c**, My proposal is to, within an agentic system, leverage robust generalist foundation-model encoders whose outputs can be tokenized with domain-specific information. For example, when a user queries with a prompt, a series of LLMs can dynamically select the appropriate foundation model(s), such as 'pose estimation', which can output skeletons of animals; 'genomic', which can be the DNA (or protein) sequence; 'neural', which can be tokenized spikes; and/or 'vision–language', which can be video captioning and actions (that is, a higher level abstraction of behavior). In this example, the assumption is that there is a link between the genome, the animal phenotype and the neural code. Critically, my proposal is to then have an LLM-based agentic system oversee accuracy per encoder and pass forward only reliable outputs for joint latent-space optimization and joint decoder. In this way, the larger agentic system need not be taken offline, but rather can be systematically updated when errors are detected in the outputs of various encoders. **d**, Building on ref. 102, such agentic systems can leverage foundation models and advances in ML and deep learning to move beyond the emergence of functionality and into adaptive intelligent systems. Genetic and neural icons are adapted from SciDraw.io; mouse images are provided by the Allen Institute and courtesy of the author of this Perspective. RAG, retrieval-augmented generation; DB, database; est., estimation; lang., language; seq, sequencing.

interconnected foundation models that can then be adapted. Namely, just as all sensorimotor information gets converted to spikes—a shared computing space that is an efficient digitization of analog signals—this information is computed within specific brain regions. Just as a cortical column in the primary visual cortex processes a particular region of the visual field and has specific coding properties related to visual features (such as texture or edges), the cortical circuit in the sensorimotor region computes in egocentric 3D kinematic, force or even muscle space[46,48].

Moreover, across the animal kingdom, we see specialization in brains everywhere, from the remarkably high resolution of hawk vision to bat echolocation and the twelve cones of the mantis shrimp's visual system. The neurons are not sitting in a single tangled mess of weights and nodes (akin to a single foundation model), but rather cleverly interconnected into neural circuits that have specific tuning to collectively solve many tasks. Perhaps we should draw inspiration from these specialized systems rather than aiming for a single model that could achieve artificial superhuman intelligence. Thus, I argue we should focus on building smart, adaptive agentic systems of specialized models.

AI agents for neuroscience

Emerging work suggests that it will soon be possible to build AI agents (agentic systems) imbued with adaptive algorithms inspired by biologically intelligent systems. Agentic systems are defined as systems imbued with some degree of autonomy, namely, they can make 'decisions' and execute code without constant human input (https://blogs.nvidia.com/blog/what-is-agentic-ai/). Agents that act as 'operating systems' are one promising avenue. They can be a series of LLMs (which is currently being explored[80,99]), but also a series of VLMs and other multimodal language models[108], which work together to ingest data, select the appropriate task-specific encoder, and link these together with downstream processing scripts. Concretely, an example of this is video behavioral analysis, where an agentic system must select the

best pose estimation model for the animal type in the video, perform semantic segmentation with another model, and then perform task programming to create ethograms[80]. To develop more adaptive agentic systems, domain-specific encoder models are essential to achieve optimal performance.

Following the example above, an agentic system I propose has an architecture based on a series of encoder models, augmented by two new features (Fig. 4c). First, the agentic system, which relies on a series of encoder modules—one for extracting poses, another for ingesting text plus images (VLMs), and one for extracting dynamics from neural spikes—is mediated by prediction errors from a decoder. Second, the outputs of each specialist encoder are jointly optimized in a latent space, which can then learn joint representations that may be inherently more brain-like and more powerful than a single foundation model alone. This approach is domain-agnostic, drawing on neuroscience to enhance adaptability.

How could this work? Encoders would be individually pretrained (think domain-specific foundation models, such as an encoder for animal behavior[91] or unified neural region-specific pretrained encoders[79,84]) and their outputs are jointly optimized into another transformer (Fig. 4c). The issue becomes that, while each encoder model starts as an expert, over time they need to learn more information. How to sense that they need to undergo learning, and how to adapt them? I propose that we could dynamically 'lock' encoder models that show robustness on OOD samples and robustness to adversarial attacks, implying that they continually need to be monitored by a dedicated LLM node to provide trustworthy outputs. Therefore, these encoders do not need to be trained (adapted) until they are deemed no longer robust. This process can be seen as analogous to the cortex–basal ganglia loop, which has a key role in learning and habit formation[109]. Namely, these encoders could be in states of 'skill learning' or 'habituation' (frozen) and be constantly monitored for robustness (Fig. 4c). When robustness levels fall below an acceptable threshold, then continual learning, memory replay, pseudolabeling or injection of new high-quality labeled data could be added without taking the entire agentic system offline, just the encoder or encoders that are not trustworthy. There has been a surge of exciting work on building OOD detection modules[110,111] and explainable attribution methods in time-series and images[112–116] that could make this an obtainable and attractive path forward.

We can also use a neuroscience-inspired approach by specifically adding prediction errors into the LLM decoder that monitors each encoder. Effectively, prediction errors provide a signed teaching signal when an uncertain prediction-outcome is detected. The development of predictive-coding-inspired networks already shows promise[39,117,118], and now with modern transformer architectures we could leverage prediction errors to not only do in-context learning within the encoder online, but signal which encoders need unlocked and updated—akin to how the brain must decide to online change a motor command to align with the internal model's predictions versus update an internal model[119]. Namely, a one-time error means 'in-context learn' and change your ongoing motor command, but a repetitive error means update your internal model (Fig. 1c). If the incoming perturbation cannot be solved with online adaptation, a new internal model would then need to be created, which takes more time and can be completed 'offline' in the unlocked model. Then methods such as continual learning, pseudolabeling and memory replay, or synaptic intelligence could be used to adapt the model online.

Moreover, discoveries in neuroscience should inspire new online learning techniques. The neural schema in the prefrontal cortex discovered by ref. 28 could inspire new approaches for adaptive memory replay and new model building as well (Fig. 1b). Models taken offline could use standard techniques such as gradient descent to fine-tune the model on the data that produced a prediction error (and therefore a drop in robustness).

The second aspect of my proposal is to jointly optimize a downstream joint latent space that takes as inputs the output tokens from each specialist encoder. This could provide an embedding that offers a richer representation of each subtask compared to jointly optimizing on raw input data, as currently done for training foundation models. Notably, this jointly optimized space can have its own LLM agent (as I describe above), with its own prediction errors, which also facilitates human-based natural language interactions. As a result, at inference time, the model then outputs human-interpretable reasoning, thereby aiding in transparency, trustworthiness and interpretability. Collectively, I can envision such an adaptive, agentic system in neuroscience being used to automate data analysis using natural language. A user could input paired video, neural and metadata data and ask, "Is there a link between the knocked-out gene and the behavior of the mouse?" Such a query could automate the selection of an appropriate genetic database and pose estimation model, query related literature, and then generate analysis code and produce novel hypotheses (Fig. 4c,d).

Lastly, independent of the agentic system itself, taking the cellular diversity found in the brain seriously in neural networks should drive innovations in architectural and agentic system design. SNNs have excitatory and inhibitory as well as neuromodulatory units, compared to current transformers. Can we merge the power of large transformers and the scale of data animals receive, with the diverse cell types that produce neural computations (Fig. 2)? Perhaps gating mechanisms akin to transformer blocks (Fig. 4b) that add a signed (excitatory or inhibitory) tokenization and masking with biological time delays can lead to innovations in such adaptive artificial systems.

Closing remarks

As we strive to move beyond traditional AI and build truly adaptive intelligence (Fig. 4d), a key lies in integrating insights from biological systems. I argue that adaptability will emerge from incorporating neuroscience-inspired principles, particularly internal models that leverage prediction error-based updating that refines internal representations based on discrepancies between expected and observed inputs, and from the brain-like modularity that structures the system with functionally distinct yet interconnected encoders, akin to sensory and motor modules that differ in data types but share an architecture. By integrating these mechanisms, agentic systems could achieve greater robustness and generalization across tasks, making them more effective in dynamic and uncertain environments. Inherent to these ideas is the notion that adaptive agentic systems can also be embodied in robotic hardware to systematically test, in a fully observable sensorimotor system, their robustness and generalization. Ultimately, such advances therefore not only push progress in AI, but also impact neuroscience by providing a new framework and testbed for theories of the mind.

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Additional information

Correspondence should be addressed to Mackenzie Weygandt Mathis.

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